AI READY ASEAN YOUTH CHALLENGE · WRITTEN PROPOSAL · MARCH 2026

EEG Motor Imagery Transformer for Hand Movement

An Open-Source BCI Research System for Motor Imagery Decoding in ASEAN Populations

PROJECT	**EEG Motor Imagery Transformer for Hand Movement**
REPOSITORY	**github.com/LaySopanha/EEG-Motor-Imagery-Transformer-for-Hand-Movement**
GEOGRAPHIC SCOPE	Cambodia (pilot study) → ASEAN (scale)
TEAM LEAD	**Lay Sopanha** · sopanha.lay@student.cadt.edu.kh
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PRIMARY **Scientific Progress** - Accelerating scientific discovery to address society's biggest challenges **SUPPORTING** Stronger Communities · Knowledge, Skills & Learning

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1. PROBLEM STATEMENT

Across Southeast Asia, millions live with destroyed voluntary motor control - through stroke, landmine injuries, spinal cord trauma, ALS, and cerebral palsy - while retaining full cognitive function. Cambodia bears the heaviest burden: over **40,000 landmine amputees** (CMAC, highest per-capita globally), **19,000 new stroke patients annually** (WHO, 2023), and an estimated **1.47M people** with motor disability from a population of just 17M (8.6% rate vs 5-6% ASEAN average). Across the region, **90+ million** people live with disabling limb conditions.

Figure 1a - Projected Productivity Loss, Cambodia (USD M)

Two scenarios 2025-2035. Sources: World Bank 2021, UN DESA 2022, UNFPA 2023

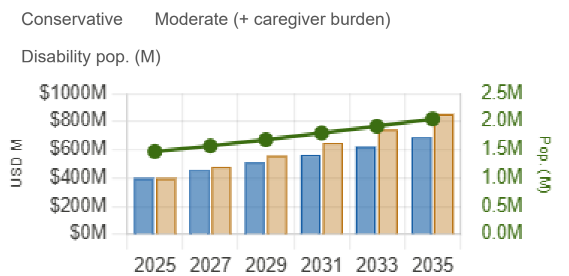
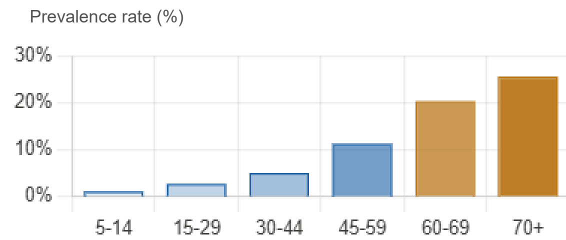


Figure 1b - Disability Prevalence by Age, Cambodia (%)

Reaches 25.6% for those over 60 - the fastest-growing cohort. Source: UNPRPD



The problem is accelerating. Some **4-6 million unexploded landmines** remain in Cambodian soil (CMAC, 2023). The aging population will drive disability prevalence to 25.6% among the 60+ cohort, growing from 9.8% to 12.5% of total population by 2035. Current annual productivity loss is **USD 400M**, projected to reach **USD 700-850M by 2035**. **50% of older Cambodians with disability have no paying work** (UNDP, 2022); 80% live where no rehabilitation specialist ever reaches them.

2. CURRENT SOLUTIONS AND THEIR GAPS

Approach	Key Limitations for ASEAN
Myoelectric (EMG) prosthetics	Require intact residual muscle - ineffective for paralysis or ALS. Cost USD 20,000-100,000; unaffordable at scale in Cambodia.
Manual rehabilitation therapy	Therapist-to-patient ratio exceeds 1:500. USD 20-50/session. Cannot reach rural populations (80% of affected people).
Academic BCI systems	Validated on fewer than 20 Western subjects. USD 50,000+ hardware, often surgical. No ASEAN-validated, open-source system exists.
Consumer neurofeedback	Designed for wellness, not motor control. Motor imagery accuracy below 65% - insufficient for reliable device control.

3. RESEARCH STAKEHOLDERS

Group	Composition and Research Role
Research subjects	Landmine amputees, stroke survivors, spinal cord injury, ALS, and cerebral palsy patients (ages 15-70, rural and urban Cambodia). Volunteer subjects for live EEG data collection; Khmer-language interface; CADT IRB approval and informed consent required.
Research partners	CDPO, Exceed Worldwide, CBM Global, rehabilitation clinics, **IDRI (CADT)**, DGIST Korea, Alt-Access (EU/IMS/Impact Hub). IDRI provides institutional research home; DGIST provides international validation.
Institutional alignment	MoSAVY, CMAC, MPTC, WHO SEARO. CADT operates under MPTC, providing direct government alignment with Cambodia's Digital Economy Master Plan 2035.

4. THE RESEARCH: EEG MOTOR IMAGERY TRANSFORMER FOR HAND MOVEMENT

This system decodes motor imagery - electrical brain activity when a person imagines moving their hand - from a non-invasive EEG headset into a real-time device command with no physical movement required. The scientific contribution is the **first validated, open-source BCI designed and tested at ASEAN scale**: prior systems used fewer than 20 Western subjects in closed codebases. This directly advances Google.org's **Scientific Progress** focus by opening a field that has systematically excluded the populations who need it most.

4.2 How It Works - From Thought to Command

FIGURE 3 — EEG MOTOR IMAGERY TRANSFORMER: END-TO-END RESEARCH PIPELINE

STEP 01 — Imagine
Person imagines moving left or right hand. No physical movement required. Motor cortex emits distinct electrical patterns for each intended direction.
MOTOR CORTEX

→

STEP 02 — Capture
EEG headset records 64-channel scalp activity at 160 Hz over a 4-second window. Signal bandpass filtered to 8-30 Hz (Mu/Beta motor rhythms).
64 CH · 160 HZ · 4 S

→

STEP 03 — Decode
HybridBrainTransformer: CNN extracts spatial features across electrodes; Transformer encoder captures long-range temporal dependencies. 528K parameters, Bayesian-tuned.
CNN + TRANSFORMER · 528K

→

STEP 04 — Act
Left/Right command sent to any connected device in under 50ms. Prosthetic, wheelchair, or keyboard - no hardware changes required at the receiving end.
<50MS · ANY DEVICE

Input	Filter	Model	Accuracy	Latency
641 steps × 64 ch	8-30 Hz Mu/Beta	CNN + 2-layer Transformer	97.37% (109 subjects)	<50ms CPU

4.3 Why AI Is Necessary - Event-Related Desynchronisation

EEG signals are in microvolts, contaminated by noise, and highly individual-specific. The key phenomenon is **Event-Related Desynchronisation (ERD)**: imagining left-hand movement drops Mu/Beta power in right-hemisphere channels (C4, FC4), and vice versa (Figure 4). This asymmetry is subtle and subject-variable - classical methods reach only ~72% accuracy. Deep learning modelling all 64 channels and 641 time-steps simultaneously achieves **97.37%**.

Figure 4 - EEG Mu/Beta Band Power During Motor Imagery: Left vs Right Hand

Event-Related Desynchronisation (ERD): power drops contralateral to the imagined movement. Left hand → right hemisphere drops (C4, FC4). Right hand → left hemisphere drops (C3, FC3). This is what the AI learns to classify. Illustrative based on Pfurtscheller & Lopes da Silva (1999).

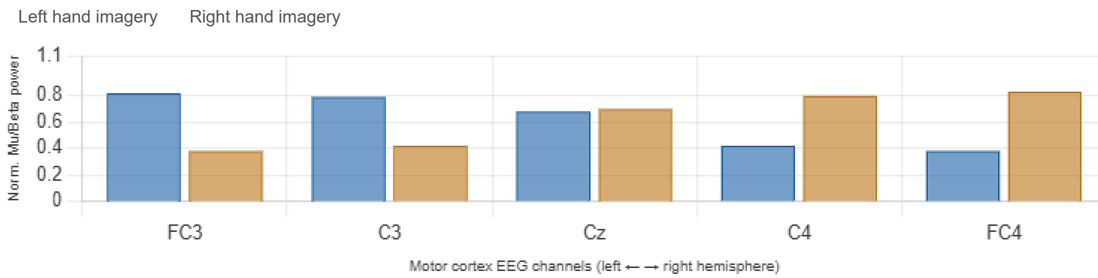
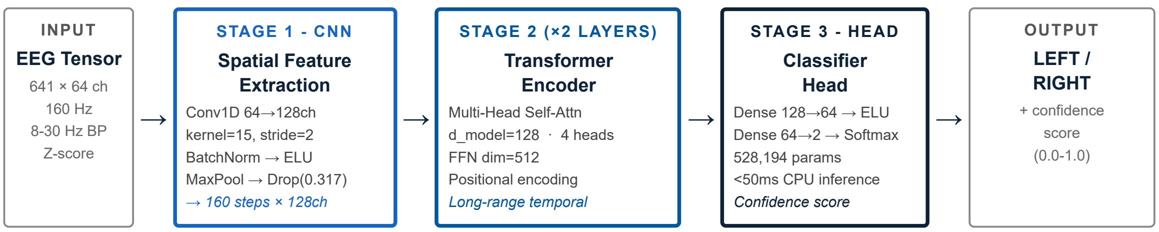


Figure 4 - Normalised Mu/Beta power (1.0 = resting baseline, lower values indicate ERD/desynchronisation). The hemispheric asymmetry is the discriminative feature; its subtlety and inter-subject variability is why deep learning outperforms classical methods.

4.4 Core Architecture: HybridBrainTransformer

FIGURE 5 — HYBRIDBRAINTRANSFORMER ARCHITECTURE · 528,194 PARAMETERS · PYTORCH · GITHUB



5. SCIENTIFIC CONTRIBUTIONS AND RESULTS

- **109-subject cross-subject validation.** Full PhysioNet dataset, no exclusions - the most comprehensively validated open BCI for this task. Prior work rarely exceeds 20 subjects, critically limiting generalisability.
- **Hybrid CNN-Transformer + Bayesian tuning.** CNN captures spatial electrode patterns; Transformer captures long-range temporal dependencies across the 4-second trial. Optuna TPE (15 trials) found optimal lr=0.000517, dropout=0.317. 3x augmentation (Gaussian noise + temporal shift) expands training from 4,898 to 14,694 samples.

Figure 6 - Classification Accuracy vs Prior Work (%)

PhysioNet EEG Motor Imagery Dataset, left vs right hand. EEG Motor Imagery Transformer highlighted.

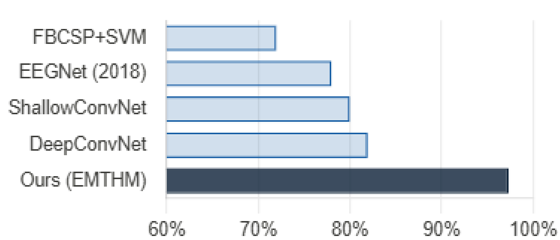
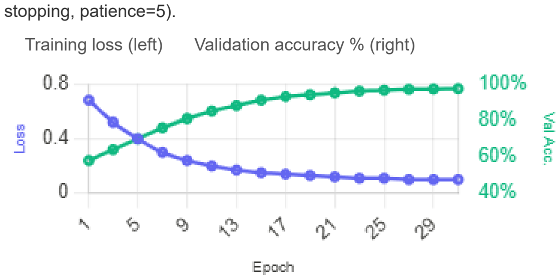


Figure 7 - Training Progress: Loss and Validation Accuracy

Full 109-subject training run. Model converged at epoch ~30 (early stopping, patience=5).



97.37%	0.97	0.98	0.98	<50ms
Overall accuracy 109 subjects	Weighted F1 both classes	Recall Left Hand	Precision Right Hand	Inference latency (CPU)

6. FEASIBILITY

Area	Detail
System status	Already built and working. Trained on NVIDIA L4 (Lightning AI) in under 90 minutes for all 109 subjects. Retraining on Cambodian data: USD 10-50/run. Runs on standard CPU at <50ms - no GPU needed at deployment.
Team	**Lay Sopanha** - research internships at IDRI (CADT) and DGIST Korea; published Transformer paper (ACET 2025); deployed HPC Kubernetes cluster at CADT. Outreach capacity proven through **Alt-Access** - a digital accessibility campaign **co-founded by Lay Sopanha** (EU/IMS/Impact Hub, 31,125 reach, 4,027 engagements).
Cost & Ethics	EEG headset ~USD 500 (OpenBCI Cyton). CADT IRB approval + informed consent required before any human data collection. EEG carries no biometric identity. Open codebase enables independent audit.

7. RESEARCH ROADMAP AND IMPACT

7.1 Development Roadmap

Phase	Timeline	Research Deliverables
Phase 1 - Complete	Done	97.37% accuracy on full 109-subject PhysioNet dataset. Automated Optuna tuning. Open-source PyTorch codebase at GitHub. Cloud GPU training pipeline operational.
Phase 2	Next 12 months	CADT IRB approval. 20-50 volunteer subjects for live Cambodian EEG data collection - the first open motor imagery dataset from Southeast Asian subjects. Cross-cultural validation results submitted for peer review.
Phase 3	Years 2-3	Cambodian EEG dataset published open-access. 2-3 rehabilitation clinic deployments through NGO partnerships. Khmer-language interface. Target: IEEE TNSRE or ICASSP publication. SaaS/API research access layer for other ASEAN institutions.

7.2 Research Impact and Sustainability

Scientific: First open EEG motor imagery dataset from Southeast Asian subjects. Reproducible methodology enabling ASEAN BCI research. Target: IEEE TNSRE publication. **Social:** Projected 50,000 users by Year 5 with USD 25M in restored annual productivity - against Cambodia's USD 700-850M projected annual loss by 2035. Aligned with SDG 3, 9, 10, and Cambodia's Digital Economy Master Plan 2035. **Sustainability:** Continues as Master's thesis at CADT/DGIST. Funding pipeline: TSC/MPTC, EU/IMS grant network (through our Alt-Access co-founder relationship), CBM Global/CMAC donor networks.

8. OUTREACH PLAN

Our strategy is built on **Alt-Access** - a digital accessibility campaign **co-founded by our team lead Lay Sopanha**, backed by the EU, IMS, and Impact Hub (2025-26). We have direct hands-on experience running every channel below. This campaign is a continuation, reframed as *"From Digital Accessibility to Neural Accessibility."* Full evidence at altaccess.site.

31,125	30,105	11,344	4,027	30
Total campaign reach	Cambodian audience	Video views	Post engagements	Live workshop attendees

Channel	Proven — Our Alt-Access Campaign (altaccess.site)	EEG Motor Imagery Transformer Plan
Video Content	Short-form YouTube/TikTok explainers in Khmer. **11,344 views** accumulated across the campaign.	"EEG Motor Imagery Transformer in 60 seconds" Khmer explainer + live EEG demo recordings. Target: 10,000+ views.
Social Media & Posts	13 posts, **4,027 engagements**. Reached tech students, tech teachers, popular Cambodian tech creators, disability community, and caregivers. 96.7% Cambodian audience.	Reactivate same segments at CADT, RUPP, ITC. Amplify via Cambodian tech creators. Disability community via CDPO/CMAC networks. Goal: surpass 4,027 baseline.
Workshop	Hands-on workshop at **Impact Hub Phnom Penh (Feb 2026)**. **30 live attendees** - students, NGO staff, community members.	"Decode Your Brain" live EEG session at CADT - participants see their own motor imagery classified in real time. Target: 50+ attendees; extend to 2 partner universities within 6 months.
Expo & Government	Physical booth at **Government Digital Expo**. Presented directly to **Minister of MPTC** - establishing government-level visibility.	Present at next MPTC/CADT expo. Leverage existing ministerial relationship. Seek MoSAVY and CMAC co-endorsement aligned with Digital Economy Master Plan 2035.